

University of Toronto

PHYB10 Intermediate Physics Laboratory I

Final Project: Developing Light Intensity Variations By Using a LED Photometer Under Different Cloud Coverage

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1.0 Abstract

This experiment investigates the impact of varying lighting conditions on the voltage output of an LED photometer. Key findings reveal a direct relationship between cloud coverage and voltage output, with reduced voltages under heavy clouds and higher outputs under sunny conditions. The result of our experiment shows that the red LED exhibited greater sensitivity to light intensity changes compared to the blue LED. Despite challenges posed by unpredictable weather and assumptions such as constant temperature and solar angles. The experiment demonstrates that low-cost photometers can effectively measure light intensity variations. Furthermore, our team conducted a literature review in related fields to support and inform subsequent analysis.

2.0 Introduction

The study of light intensity and its environmental variations has numerous applications, ranging from environmental monitoring to optical engineering. This project focuses on analyzing the relationship between changing lighting conditions and voltage output using a photometer constructed with simple yet effective components.

The experimental setup employs red and blue light-emitting diodes (LEDs) as light detectors. To enhance the accuracy of the measurements, a 3D-printed photometer was utilized to minimize interference from stray light, while an Arduino microcontroller recorded the output under various lighting conditions, including heavy clouds, light clouds, and sunny days.

The primary objective of our experiment is to assess the sensitivity of red and blue LEDs under varying lighting conditions. Additionally, our team conducted research in related fields to support and inform further analysis.

3.0 Method

3.1 Assumptions

To simplify the experiment and focus on analyzing the relationship between lighting conditions and the voltage output of the photometer, the following assumptions were made:

- **Constant Temperature:** Based on the experimental principle of controlling variables, we assume the ambient temperature during the experiment is constant. This approach helps us prevent the potential effects of temperature variations on LED sensitivity and circuit performance.
- **Stable Solar Angle:** The solar angle for noon on Nov. 24th was 25.3 degrees, and Nov. 28th was 24.8 degrees [1]. We assume that the changes in the angle of the sun relative to our equipment are negligible. This ensures a constant solar radiation intensity and a constant light source distance.
- **Definitions of Cloud Coverage Levels:**
    1. Heavy Cloudy: The sun is covered by clouds, with little to no direct sunlight visible.
    2. Light Cloudy: there is partial sunlight to pass through.
    3. Sunny: Direct sunlight fully illuminating the area.
- **Uniform Sunlight Intensity:** Sunlight intensity across the experimental field of view is assumed to be uniform, without significant localized variations.

- **Constant Power Supply:** We assume the power supply for our measurement equipment is constant.

3.2 Equipment

This experiment uses a standard LED photometer circuit to measure voltage changes under varying lighting conditions. Key features of the equipment include:

- **Resistors:** Limit current flow to protect LEDs and ensure stable amplifier operation, directly affecting the photometer's sensitivity.
- **Amplifier (Op-Amp):** Converts small currents generated by the LEDs into measurable voltage outputs, with a feedback resistor determining the amplification gain.
- **Capacitors:** Filter noise in the voltage signal, enhancing stability and reducing fluctuations.
- **LEDs:** Red and blue LEDs detect light intensity changes and generate small currents amplified for measurement.
- **3D-Printed LED Barrier:** Blocks stray light, ensuring measurements are influenced only by light from the intended source.
- **Arduino Microcontroller:** Records and averages voltage outputs every 10 seconds using a pre-installed sketch for consistent data collection.
- **Power Supply:** A 5 V power source ensures stable operation and protects circuit components. The laptop has exactly 5 V for the power supply [2].
- **Multimeter:** Verifies circuit functionality during setup by monitoring voltage changes.

This setup, following the lab-provided schematic, ensures reliable and repeatable measurements under different lighting conditions.

3.3 Theory

This section outlines the theoretical principles and calculations underlying the experiment, focusing on the photometer's operation and the relationship between its components.

3.3.1 Mimm's Effect and LED Response

When exposed to light, LEDs generate a small photocurrent due to the Mimm's Effect, which is proportional to the light intensity at the LED's sensitive wavelength. For this experiment:

- The red LED detects light near 650 nm.
- The blue LED detects light near 470 nm.
- This photometer current(I) is amplified by the circuit to produce a measurable voltage output.

3.3.2 Amplifier Output Voltage

The operational amplifier (op-amp) in the circuit increases the LED's weak photocurrent signal. The output voltage (V_{out}) is given by: $V_{out} = IA$, where A is the amplification factor determined by the feedback resistor (R_f): $A = 1 + \frac{R_f}{R_{in}}$, where R_{in} is the input resistance, ensuring linearity of the voltage response.

3.3.3 Capacitor Filtering

Capacitors in the circuit smooth out voltage fluctuations caused by noise or rapid light intensity changes. The effectiveness of the filtering depends on the capacitor value(C) and resistance(R) in the circuit: $V_{filtered} = V_{input} * \frac{1}{(1+j2\pi fRC)}$, where f is the signal frequency. This ensures that only stable signals contribute to the recorded voltage.

3.3.4 Voltage Averaging

To improve data reliability, the Arduino program calculates the average voltage(V) over a set period(2 seconds):$V_{avg} = \frac{V_1+V_2}{2}$, where V_1 and V_2 represents voltage on the first second and second second.

3.3.5 Impact of Light and Cloud Conditions

Light intensity decreases as cloud coverage increases due to scattering and absorption. This relationship is captured by the photometer as a reduction in voltage output: $I_{cloudy} < I_{sunny}\ and\ V_{out,cloudy} < V_{out,sunny}$, where I_{cloudy} and $V_{out,cloudy}$ represent the photometer current and voltage under cloudy conditions.

3.4 Procedure

3.4.1 Circuits Building

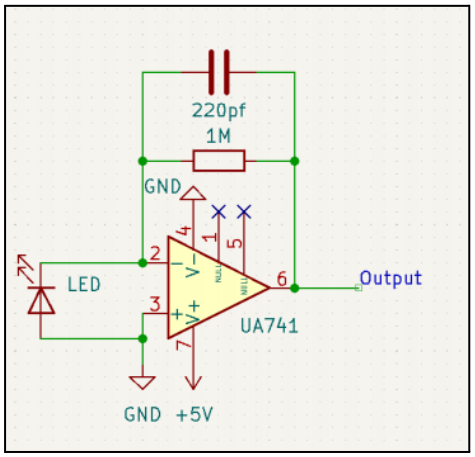


Figure 3.4.1: Circuits Diagram

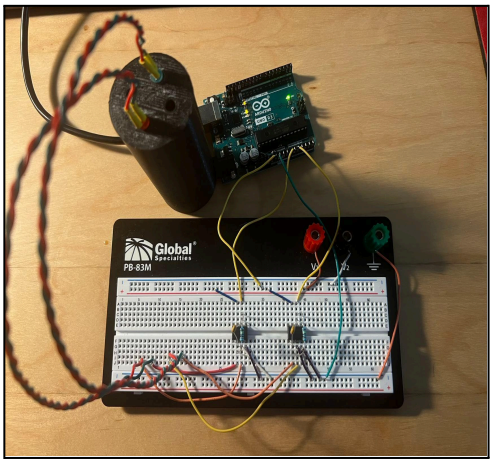


Figure 3.4.2: Photometer

The two figures above illustrate our photometer setup. Figure 3.4.1 shows the detailed circuits for both the blue and red LEDs. The green board represents the Arduino UNO microcontroller, while the black cylinder functions as the photometer barrier.

The experimental procedure involved several steps.

- The photometer barrier was fabricated using a 3D printer
- The circuit depicted in Figure 3.4.1 was assembled, and the functionality of the LEDs was verified.
- To ensure the portability of the detector, we welded the unit.
- The Arduino UNO board was connected to a laptop, and measurements were taken under real-life conditions. In our experiment, we faced the detector to the sun and plugged the board into the laptop as a power supply.

3.5 Data Collection

Arduino UNO board and the Arduino app are used to collect experiment data. We used the codes and the app provided on Quercus for data generation and collection.

4.0 Result Analysis

In this section, we present the results of our experiments and analyze the findings. For our experiment, two types of measurements were performed: First, we focused on the effect of different lighting conditions on the outputs of the two LEDs and took several targeted measurements under different lighting conditions; Second, to better visualize the effect of changing lighting conditions on the voltages caused by variations in weather, we conducted a single prolonged measurement that captures 3 different lighting conditions.

The two experiments were both taken on November 24, 2024, starting at 12:30 PM, over one hour. The ambient temperature during the experiment was 7 degrees Celsius.

4.1 Base Condition for Testing LED Output

Figures 4.1.1 and 4.1.2 are the result of the LEDs under the darkroom condition. The blue LED showed a voltage of 1.92 volts and the red LED showed a voltage of 1.92 to 1.93 volts.

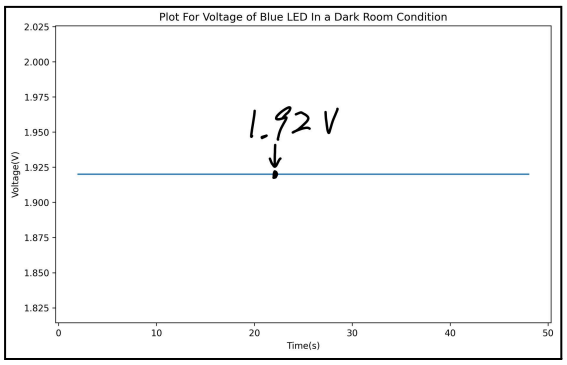


Figure 4.1.1: Voltage of blue LED in dark room

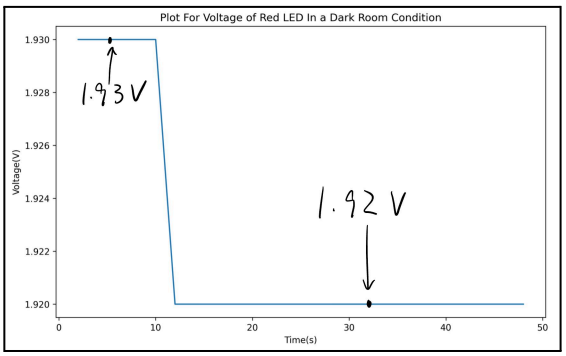


Figure 4.1.2: Voltage of red LED in dark room

4.2 Condition 1: Outside-Dark Condition

The two graphs below show the output for both of our LED lights:

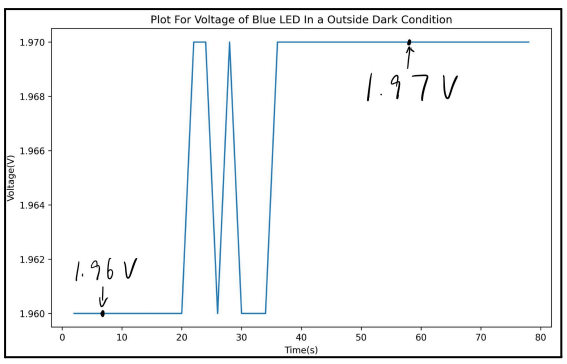


Figure 4.1.3: Voltage of blue LED in outside dark condition

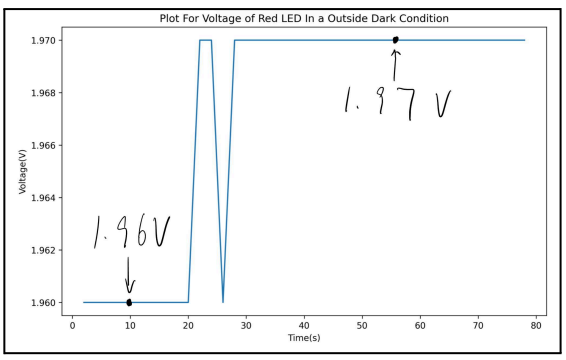


Figure 4.1.4: Voltage of red LED in outside dark condition

To simulate dark conditions, we inverted the photometer to shield it from ambient light. Under these circumstances, both the blue and red LEDs exhibited voltages ranging from 1.96 to 1.97 volts.

4.3 Condition 2: Heavy Cloudy Condition

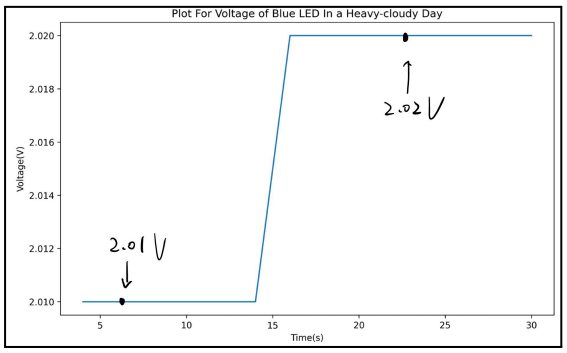


Figure 4.1.5: Voltage of blue LED in a heavy cloudy condition, 12:28 PM

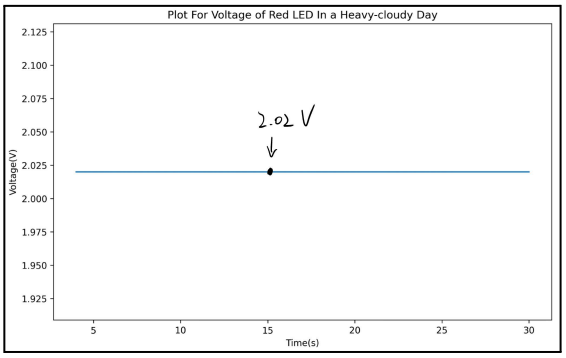


Figure 4.1.6: Voltage of red LED in a heavy cloudy condition, 12:28 PM

For the Heavy-Cloud condition, the blue LED showed a voltage of 2.01 to 2.02 volts and the red LED showed a voltage of 2.02 volts.

4.4 Condition 3: Light Cloudy Condition

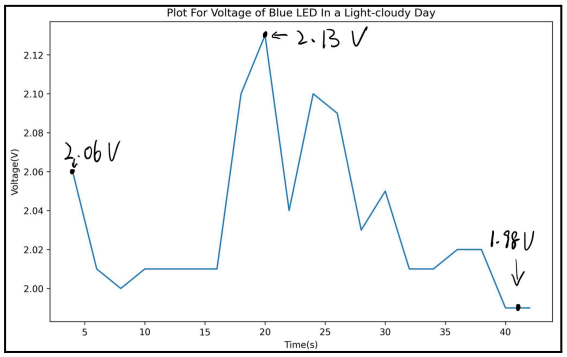


Figure 4.1.7: Voltage of blue LED in a light cloudy condition, 12:22 PM

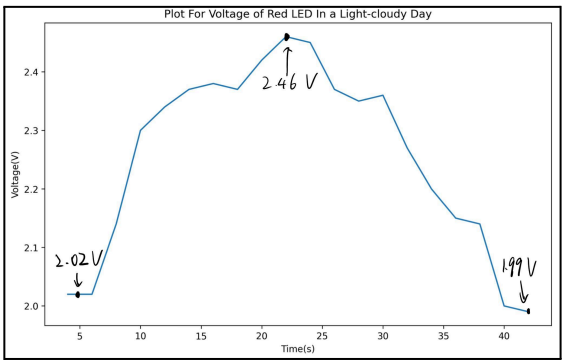


Figure 4.1.8: Voltage of red LED in a light cloudy condition, 12:22 PM

For the light cloud condition, the blue LED showed a voltage of 1.98 to 2.13 volts and the red LED showed a voltage of 1.99 to 2.46 volts.

4.5 Condition 4: Sunny Condition

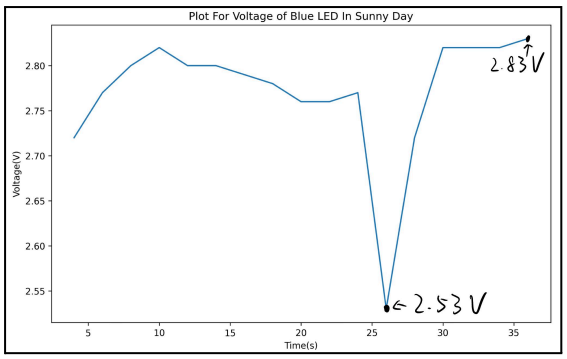


Figure 4.1.9: Voltage of blue LED in a sunny condition, 13:15 PM

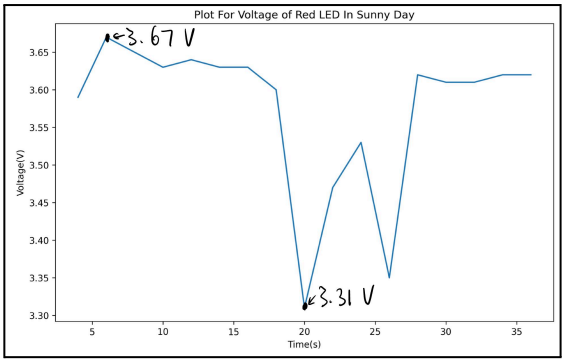


Figure 4.1.10: Voltage of red LED in a sunny condition, 13:15 PM

For the sunny conditions, the blue LED showed a voltage of 2.53 to 2.83 volts and the red LED showed a voltage of 3.31 to 3.67 volts. Below is a table showing the average values of measurement under different conditions.

Table 4.1.1: Data for average voltage and the voltage change for different conditions

	Average Blue	**Average Red**	$\Delta V_{blue}Room$	$\Delta V_{red}Room$	$\Delta V_{blue}Outside$	$\Delta V_{red}Outside$
Room	1.92	1.922	0	0	-0.046	-0.045
Outside Dark	1.966	1.967	0.046	0.045	0	0
Heavy Cloudy	2.016	2.02	0.096	0.098	0.05	0.053
Light Cloudy	2.035	2.255	0.115	0.333	0.069	0.288
Sunny	2.771	3.575	0.851	1.653	0.805	1.608

4.6 Long-Term Measurement

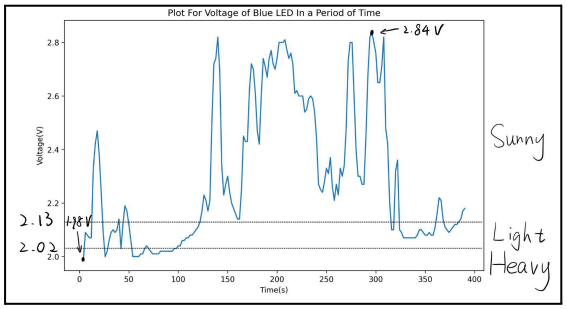


Figure 4.2.1: Voltage of blue LED in a period of time, 13:25 PM

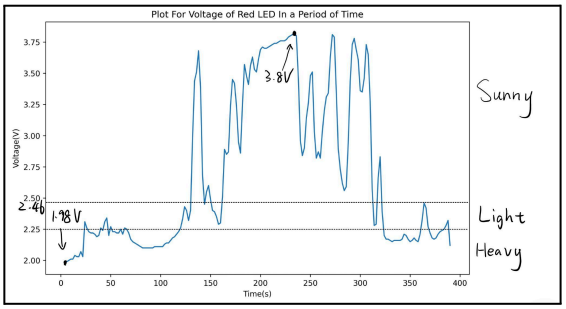


Figure 4.2.2:Voltage of red LED in a period of time, 13:25 PM

The two figures above show the result of our 6-minute measurement with three distinct conditions: heavy cloudy, light cloudy, and sunny. The heavy cloudy condition is characterized by voltages below 2.02 V for the blue LED and below 2.25 V for the red LED.

The light cloudy condition corresponds to voltages between 2.02 and 2.13 V for the blue LED, and between 2.25 and 2.46 V for the red LED. Measurements exceeding 2.13 V for the blue LED and 2.46 V for the red LED indicate sunny conditions. The maximum recorded voltages during the measurement period were 2.84 V for the blue LED and 3.8 V for the red LED.

Here is a link to a short video for this 6-minute measurement period with 3 different conditions: https://www.youtube.com/watch?v=xAKgnK2-xOk

4.7 Comparative Analysis

Based on our experiment result, the voltage readings for the two LEDs are similar under different conditions. However, the red LED exhibits a significantly greater change in voltage compared to the blue LED under light cloudy and sunny conditions.

5.0 Discussion

5.1 Strengths

5.1.1 Simple Equipment for Reliable Measurements

This experiment used a simple LED-based design. Although the equipment is simple, this setup effectively captured distinguishable light emission peaks under varying light conditions. The system makes it suitable for conducting basic planar optics experiments.

5.1.2 High Repeatability

Considerable number of errors are eliminated by automated data collection through an Arduino program, the experimental results are easily generated.

5.2 Challenging

5.2.1 Unpredictable Weather Conditions

Toronto's rapidly changing weather created significant challenges for data collection. Rainstorms disrupted multiple attempts to measure under clear skies, the measurements can’t have the ideal conditions.

5.2.2 Human Factor

During the experiment, the photometer was positioned to directly face the Sun. Any deviation from this alignment could result in the barrier casting a shadow, thereby introducing errors in the recorded data.

5.2.3 Control Variables

Assumptions like constant temperature and stable solar angles simplified analysis but introduced potential errors:

1. Slight shifts in the Sun's position affect light intensity and angle.
2. Temperature fluctuations may have altered the LEDs' sensitivity.

5.3 Explanation of Observed Results

5.3.1 Cloud Coverage and Light Intensity

Clouds scatter and absorb sunlight, reducing direct light reaching the photometer. Thick clouds diminished voltage output significantly, while clear skies allowed maximum light transmission.

5.3.2 Effective Experimental Design
The amplifier circuit amplified small LED currents, ensuring even subtle variations in light intensity were measurable. Moreover, the LED barrier minimized stray light interference.

5.3.3 Impact of Assumptions
- Constant temperature and solar angle: Short measurement periods justified ignoring temperature and solar angle effects on LED sensitivity.
- Negligible hand tremors: Minor manual alignment inconsistencies were insignificant in affecting overall trends.

5.4 Suggested Improvements and Next Steps
5.4.1 Enhancing Measurement Reliability
The reliability of future testing could be improved by the inclusion of a temperature sensor that would help compensate for the influence of ambient temperature variation on LED sensitivity.

5.4.2 Expanding the Experiment Scope
The experiment could use other different wavelength LEDs to be the photometer. Furthermore, extending measurements to various times of the day such as the sunrise, midday and sunset would enable a closer look at how solar angle affects light intensity.

6.0 Conclusion and Further Analysis
6.1 Conclusion
The research proved the effect of light conditions on the voltage output of an LED photometer.
For the three distinct light conditions: heavy cloudy, light cloudy and sunny.
- The voltage has a clear negative relation with the coverage of the cloud.
- Compared with the base condition the more direct sunlight goes through the cloud the greater the change of voltage.
- The red LED has more sensitivity to light compared with the blue LED.

In spite of the difficulties caused by the varying weather conditions of Toronto, the reasonable assumptions and the advanced design of the experiment made it possible to observe quite a number of data trends. The results obtained seem to indicate that the photometer as a cheap measurement device can be useful in environmental monitoring and optical studies.

6.2 Further Analysis
6.2.1 Automatic Spraying Sun Photometer
A study highlighted the design of a low-cost sun photometer for measuring aerosol optical depth (AOD), incorporating algorithms to address calibration issues [3]. This reinforces the potential of affordable devices for large-scale atmospheric research [3].

6.2.2 Advanced measurements of gas scattering and absorption
Research on gases like carbon dioxide emphasized the importance of calibration with wider wavelength coverage to improve atmospheric measurement accuracy [4].

6.2.3 Automated Neutron Activation Photometer AOD measurements

A study on sun photometers showed how temperature compensation improves data accuracy, highlighting the value of integrating temperature control into optical devices [5].

These cases illustrate the fact that the accuracy of measurement of natural light can be improved through the progress of instrumentation, calibration and temperature control. Implementing such thoughts in future studies would also most likely improve the accuracy and range of the obtained results.

Reference

[1]. Gyorfi, S. (2020). *Sun Position and Path* (Version 1.0) [Mobile app]. Apple App Store. https://apps.apple.com/us/app/sun-position-and-path/id1502500167

[2] Lenovo. (n.d.). *Lenovo Legion 9 16 User Guide: Product specifications* [Manual]. Lenovo. https://download.lenovo.com/pccbbs/pubs/legion_9_16_irx9/user_guide/en/help_product_specifications.html

[3] Ma, J., Wu, X., Chen, X., Wu, L., Zhang, Z., & Zhang, W. (2021). Automated low-cost LED-based sun photometer for aerosol optical depth measurements at city scale. Remote Sensing, 13(22), 4585. https://doi.org/10.3390/rs13224585

[4] Langridge, J. M., Laurila, T., Wooldridge, P. J., & Cohen, R. C. (2021). Scattering and absorption cross sections of atmospheric gases in the ultraviolet–visible wavelength range (307–725 nm). Atmospheric Chemistry and Physics, 21(14), 14927–14942. https://doi.org/10.5194/acp-21-14927-2021

[5] Liu, J., Liu, W., & Li, Y. (2021). Calibration of automatic sun photometer with temperature compensation for aerosol optical depth measurements. Remote Sensing, 14(1), 66. https://doi.org/10.3390/rs14010066